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Study of the suitability of two ammonium-based ionic liquids for the extraction of benzene from its mixtures with aliphatic hydrocarbons

Patricia F. Requejo, Noelia Calvar, Elena Gómez, Ángeles Domínguez*

Advanced Separation Processes Group, Department of Chemical Engineering, University of Vigo, Campus Lagoas-Marcosende, 36310 Vigo, Spain

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The ammonium-based ionic liquids butyltrimethylammonium bis(trifluoromethylsulfonyl)imide, $[N_{4111}][NTf_2]$ and tributylmethylammonium bis(trifluoromethylsulfonyl)imide, $[N_{4441}][NTf_2]$ have been studied as separation agents on the extraction of benzene from its mixtures with the long chain aliphatic hydrocarbons octane, decane and dodecane. The liquid–liquid equilibria of the ternary systems {octane (1) + benzene (2) + $[N_{4111}][NTf_2]$ (3)}, {decane (1) + benzene (2) + $[N_{4111}][NTf_2]$ (3)}, {dodecane (1) + benzene (2) + $[N_{4111}][NTf_2]$ (3)} and {dodecane (1) + benzene (2) + $[N_{4441}][NTf_2]$ (3)} at $T = 298.15$ K and {octane (1) + benzene (2) + $[N_{4441}][NTf_2]$ (3)} at $T = (298.15, 308.15$ and $318.15)$ K and atmospheric pressure were determined. From the experimental results, an analysis of the influence of the cation of the ammonium-based ionic liquids and the alkyl chain length of the aliphatic hydrocarbons was performed; together with a study of the effect of the temperature on the liquid-liquid extraction of benzene. The extraction effectiveness was evaluated by the calculation of solute distribution ratio and selectivity derived from the tie-lines compositions. Besides, a comparison of the obtained results with those found in literature for systems with sulfolane, which is the most industrially used solvent, was performed. Finally, the experimental tie-lines were correlated using the NRTL thermodynamic model.

1. Introduction

The increasing concern about reducing the environmental impact of the industrial processes over the last years has led to the development of novel processes based on greener technologies [1]. Nowadays, research is focused on decreasing the process costs and minimizing the environmental damages caused by traditional organic solvents used in many industrial processes.

In the petrochemical industry, the separation of mixtures of aromatic and aliphatic hydrocarbons is challenging since these hydrocarbons have boiling points in a close range and several combinations form azeotropic mixtures. The conventional processes employed industrially for recovering aromatic hydrocarbons include the use of traditional organic solvents such as sulfolane, *N*-methyl pyrrolidone (NMP) or carbonates among others [2–7]. This kind of solvents are flammable, volatile and toxic, and imply high energy costs associated to their recovery; therefore, the substitution of these solvents by new and eco-friendly alternatives has increasingly been investigated over the past decade.

In this context, ionic liquids (ILs) have garnered a great attention as a sustainable alternative to replace the current separation agents used in the industry because of their attractive properties, such as their negligible vapour pressure, their thermal and chemical stability and the possibility of tailoring their physical and chemical properties for a specific application by the combination of cation and anions.

In this paper, a study about the application of two ionic liquids based on ammonium cation in the separation of benzene from its mixtures with long chain aliphatic hydrocarbons is presented. The ILs, $[N_{4111}][NTf_2]$ and $[N_{4441}][NTf_2]$, were selected since in general terms the ammonium-based ILs present lower toxicity in comparison with the ILs based on imidazolium or pyridinium cations [8–10]. Moreover, due to the fact that the aliphatic hydrocarbons present in naphta range from C_4 to C_{12} [11] and the shorter chain length hydrocarbons have been extensively studied in this kind of separation over the last years, the long chain aliphatic hydrocarbons octane, decane and dodecane were chosen to study the extraction of benzene. To our knowledge, no information about other ternary systems, which could be compared with the studied systems, has been published.

* Corresponding author.
E-mail address: admguez@uvigo.es (Á. Domínguez).

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Therefore, in order to evaluate the suitability of the selected ILs as separation agents on the extraction of benzene, the experimental determination of the liquid-liquid equilibrium (LLE) for the ternary systems {octane (1) + benzene (2) + $[N_{4111}][NTf_2]$ (3)}, {decane (1) + benzene (2) + $[N_{4111}][NTf_2]$ (3)}, {dodecane (1) + benzene (2) + $[N_{4111}][NTf_2]$(3)} and {dodecane (1) + benzene (2) + $[N_{4441}][NTf_2]$ (3)} at $T = 298.15$ K and {octane (1) + benzene (2) + $[N_{4441}][NTf_2]$ (3)} at T = (298.15, 308.15 as 318.15) K and atmospheric pressure were performed. Besides, solute distribution ratio, β, and selectivity, S, were calculated from the LLE data to evaluate the separation capability of the studied ILs. Finally, the Non-Random Two Liquids (NRTL) thermodynamic model [12] was used to correlate the experimental LLE data for the ternary mixtures studied in this work.

2. Experimental

2.1. Chemicals

The purities, suppliers and experimental and literature densities of the pure compounds used in this work at $T = 298.15$ K and atmospheric pressure together with the water content of the studied ionic liquids are summarized in Table 1 [13–16].

Benzene, octane, decane and dodecane were degassed ultrasonically and dried over molecular sieves type $3 \cdot 10^{-10}$ m (Sigma-Aldrich) without any further purification. Prior to the experiments, the ionic liquids $[N_{4111}][NTf_2]$ and $[N_{4441}][NTf_2]$ were subjected to vacuum ($p = 0.2$ Pa) at moderate temperature ($T = 343.15$ K) for at least 48 h to reduce the initial water content to negligible values. Once dried, their water content was measured using a Mettler ToledoC20 Coulometric KF Titrator with an uncertainty in the measurement of ±5%. The reactants used were Hydranal-Coulomat CG and Hydranal-Coulomat AG, supplied by Sigma-Aldrich, as cathodic and anodic titrants, respectively. The results obtained are shown in Table 1. Finally, to avoid moisture, ionic liquids were stored in bottles under an inert atmosphere in a glove box.

2.2. Apparatus and procedure

Densities of the pure compounds and mixtures were measured using an Anton Paar DSA-5000M digital vibrating-tube densimeter with an uncertainty in the measurement of the density is of $\pm 3 \cdot 10^{-3}$ g cm^{-3} for the ILs and $\pm 1 \cdot 10^{-3}$ g cm^{-3} for benzene and the aliphatic hydrocarbons taking into account the purities of the different compounds. These uncertainties were estimated considering the discussion undertaken on this issue in the work reported by Chirico et al. [17]. Moreover, the preparation (by weighing) of all the samples involved in the experimental work were carried out on a Mettler AX-205 Delta Range balance with an uncertainty of the measurement of $\pm 3 \cdot 10^{-4}$ g.

The experimental LLE data were determined at the studied temperatures (T = 298.15, 308.15 and 318.15 K) and atmospheric pressure. For the experimental determination of the tie-lines, mixtures of the studied compounds within the immiscible region were introduced into glass cells sealed with silicon covers. The mixtures were vigorously stirred for at least 5 h in order to allow an intimate contact between phases, and then allowed to settle overnight in a PoliScience thermostatic bath with digital temperature controller to ensure the complete split of the phases. The control of the temperature during the experiments was carried out using a digital thermometer ASL model F200 with an uncertainty in the measurement of ±0.01 K. Next, samples of the upper and lower phases were withdrawn using a syringe and their compositions were determined.

The LLE data for the ternary systems containing dodecane as aliphatic hydrocarbon were obtained through the determination of the tie-lines compositions using gas chromatography-mass spectrometry (GC-MS) as analysis technique, whereas for the remaining ternary systems, {octane or decane (1) + benzene (2) + $[N_{4111}][NTf_2]$ (3)} and {octane (1) + benzene (2) + $[N_{4441}][NTf_2]$ (3)}, the determination of the tie-lines compositions was carried out by means of the previous determination of the solubility curves by the "cloud point" method [18] using density measurements for obtaining the phase compositions. These two different procedures were used due to the fact that dodecane presents a very low solubility in the studied ILs, which implies that the method involving density measurements does not allow the calculation of the compositions of the tie-lines of the systems involving dodecane with enough accuracy.

The determination of the phase compositions of the ternary systems involving dodecane was carried out using a gas chromatograph (HP 5890 Series II) equipped with a HP-5MS capillary column (60 m length, 0.250 mm internal diameter, $0.25 \cdot 10^{-6}$ m film thickness) and a mass selective detector HP 5971. The GC was equipped with an injector liner filled with quartz glass to prevent the column contamination with the ionic liquid and a pre-column to collect the ionic liquid that could have not been retained by the liner. The carrier gas was helium, with a flow rate of 1 mL min^{-1} in the column. Injection was done by split-less and the injection volume was of $1 \cdot 10^{-6}$ mL. The temperature of the column was initially $T = 343.15$ K for 8 min, followed by a 55 K min^{-1} ramp up to $T = 503.15$ K and maintained for 10.6 min. The injector and detector temperatures were fixed at $T = 553.15$ K. Quantitative analysis of the phase compositions was enabled using nonane as internal standard and acetone as diluent to maintain a homogeneous mixture.

Table 1
Supplier, mass fraction purity and experimental and literature data of density (ρ) of pure compounds at T = 298.15 K and atmospheric pressure as well as water content (w_w) of the studied ionic liquids.

Compound	Supplier	Purity/(mass fraction)	ρ/(g cm^{-3})		w_w/(ppm)
			Exp.	Lit.	
$[N_{4111}][NTf_2]$	Iolitec	>0.99	1.39299	1.39307[a]	77
$[N_{4441}][NTf_2]$	Iolitec	>0.99	1.26165	1.2628[b]	68
benzene	VWR	≥0.999	0.87371	0.87360[c]	–
octane	Merck	≥0.99	0.69886	0.69862[c]	–
decane	Merck	≥0.99	0.72615	0.72616[d]	–
dodecane	Merck	≥0.99	0.74528	0.74514[d]	–

Standard uncertainties u are $u(\rho) = \pm 0.003$ g cm^{-3} (for the ILs) and ± 0.001 g cm^{-3} (for the hydrocarbons), $u(T) = \pm 0.01$ K and $u(w_w) = \pm 5\%$.
[a] From Ref. [13].
[b] From Ref. [14].
[c] From Ref. [15].
[d] From Ref. [16].

Table 2
Experimental LLE data, solute distribution ratio in molar and mass fraction, β, and selectivity in molar fraction, S, for the studied ternary systems at the studied temperatures and atmospheric pressure.

Upper-phase		Lower phase		β (in molar fraction)	β (in mass fraction)	S
x_1^{I}	x_2^{I}	x_1^{II}	x_2^{II}			
{octane (1) + benzene (2) + $[N_{4111}][NTf_2]$ (3)} at T = 298.15 K						
0.959	0.041	0.033	0.056	1.36	0.42	39.86
0.897	0.103	0.025	0.126	1.21	0.38	43.33
0.847	0.153	0.025	0.177	1.15	0.38	38.69
0.797	0.203	0.028	0.225	1.11	0.37	31.91
0.745	0.255	0.030	0.271	1.06	0.37	26.11
0.691	0.309	0.033	0.318	1.03	0.37	21.76
0.650	0.350	0.034	0.355	1.02	0.38	19.45
0.583	0.417	0.034	0.406	0.95	0.37	16.60
0.524	0.476	0.034	0.448	0.94	0.37	14.72
0.460	0.540	0.032	0.488	0.90	0.37	12.95
0.372	0.628	0.029	0.542	0.86	0.37	11.14
0.275	0.725	0.029	0.588	0.82	0.36	7.66
0.201	0.799	0.019	0.629	0.81	0.35	8.37
0.141	0.859	0.016	0.673	0.78	0.37	7.12
0.072	0.928	0.009	0.719	0.78	0.38	5.96
{decane (1) + benzene (2) + $[N_{4111}][NTf_2]$ (3)} at T = 298.15 K						
0.985	0.015	0.024	0.021	1.44	0.53	58.41
0.937	0.063	0.026	0.084	1.34	0.51	48.55
0.869	0.131	0.027	0.161	1.23	0.49	39.59
0.762	0.238	0.027	0.248	1.04	0.43	29.98
0.592	0.408	0.023	0.380	0.93	0.40	24.12
0.516	0.484	0.020	0.437	0.90	0.40	22.87
0.401	0.599	0.017	0.516	0.86	0.39	20.85
0.282	0.718	0.012	0.590	0.82	0.39	18.61
0.157	0.843	0.007	0.678	0.80	0.40	17.36
0.090	0.910	0.005	0.721	0.79	0.40	15.83
{dodecane (1) + benzene (2) + $[N_{4111}][NTf_2]$ (3)} at T = 298.15 K						
0.900	0.100	0.006	0.097	0.97	0.43	134.95
0.857	0.143	0.007	0.137	0.96	0.43	125.88
0.774	0.226	0.007	0.212	0.94	0.43	103.99
0.698	0.302	0.007	0.278	0.92	0.43	91.31
0.625	0.375	0.007	0.340	0.91	0.43	84.92
0.549	0.451	0.007	0.400	0.89	0.43	70.77
0.411	0.589	0.006	0.506	0.86	0.43	58.26
0.297	0.703	0.005	0.580	0.83	0.41	47.39
0.146	0.854	0.005	0.662	0.77	0.38	22.61
0.071	0.929	0.005	0.694	0.75	0.36	11.78
{dodecane (1) + benzene (2) + $[N_{4441}][NTf_2]$ (3)} at T = 298.15 K						
0.935	0.065	0.023	0.090	1.38	0.52	55.30
0.885	0.115	0.022	0.152	1.32	0.51	52.22
0.787	0.213	0.019	0.256	1.20	0.49	49.06
0.658	0.342	0.018	0.375	1.10	0.47	39.68
0.546	0.454	0.016	0.466	1.03	0.46	34.76
0.423	0.577	0.014	0.536	0.93	0.42	27.29
0.292	0.708	0.012	0.610	0.86	0.39	20.40
0.203	0.797	0.011	0.661	0.83	0.38	15.71
0.094	0.906	0.009	0.723	0.80	0.37	8.04
{octane (1) + benzene (2) + $[N_{4441}][NTf_2]$ (3)} at T = 298.15 K						
0.970	0.030	0.091	0.092	3.07	0.85	32.59
0.933	0.067	0.089	0.169	2.53	0.74	26.55
0.833	0.167	0.089	0.300	1.80	0.59	16.75
0.727	0.273	0.081	0.418	1.53	0.57	13.84
0.627	0.373	0.071	0.510	1.37	0.55	12.09
0.520	0.480	0.074	0.579	1.21	0.53	8.45
0.401	0.599	0.063	0.638	1.07	0.49	6.83
0.271	0.729	0.052	0.705	0.97	0.48	5.00
0.138	0.862	0.046	0.758	0.88	0.46	2.63
0.068	0.932	0.028	0.809	0.87	0.48	2.08
{octane (1) + benzene (2) + $[N_{4441}][NTf_2]$ (3)} at T = 308.15 K						
0.986	0.014	0.119	0.030	2.11	0.56	17.55
0.925	0.075	0.112	0.145	1.94	0.57	16.06
0.836	0.164	0.100	0.277	1.69	0.55	14.05
0.726	0.274	0.093	0.396	1.45	0.53	11.32
0.620	0.380	0.083	0.499	1.31	0.53	9.76
0.510	0.490	0.077	0.557	1.14	0.48	7.56
0.402	0.598	0.067	0.626	1.05	0.47	6.25
0.276	0.724	0.056	0.693	0.96	0.47	4.67
0.172	0.828	0.045	0.760	0.92	0.49	3.52
0.070	0.930	0.023	0.801	0.86	0.46	2.58

(*continued on next page*)

Table 2 (*continued*)

Upper-phase		Lower phase		β (in molar fraction)	β (in mass fraction)	S
x_1^{I}	x_2^{I}	x_1^{II}	x_2^{II}			
{octane (1) + benzene (2) + $[N_{4441}][NTf_2]$ (3)} at T = 318.15 K						
0.952	0.048	0.112	0.110	2.30	0.66	19.66
0.900	0.100	0.103	0.186	1.86	0.56	16.20
0.839	0.161	0.101	0.269	1.68	0.54	13.97
0.735	0.265	0.093	0.384	1.45	0.52	11.49
0.639	0.361	0.084	0.470	1.30	0.51	9.90
0.538	0.462	0.076	0.543	1.18	0.49	8.37
0.421	0.579	0.062	0.646	1.12	0.53	7.55
0.307	0.693	0.054	0.702	1.01	0.51	5.77
0.137	0.863	0.034	0.812	0.94	0.55	3.85
0.067	0.933	0.019	0.833	0.89	0.52	3.17

Standard uncertainties u are $u(x) = \pm 0.007$ (by "cloud point" method), $u(x) = \pm 0.004$ (by GC-MS) and $u(T) = \pm 0.01$ K.

Since ILs have negligible vapour pressure, they cannot be analysed by GC-MS, and therefore the ionic liquid composition was determined by a mass balance from the determined composition of the hydrocarbons. All the samples were measured in duplicate to obtain a mean value. In order to obtain the error of the technique, ternary mixtures of known composition were prepared by weighing. After the analyses of these samples by GC-MS, the compositions calculated were compared with those obtained by weighing and the maximum error observed was ±0.004 in mole fraction.

Moreover, the experimental procedure for the determination of the LLE data by the "cloud point" method has been described in detail in previous works [19,20]. The composition of the IL-rich phase (lower phase) was calculated using a polynomial expression of density vs. composition for each system, which are shown in Table S1 (given as Supporting Information). These expressions were obtaining from solubility curves of the corresponding ternary mixture. In the case of the aliphatic hydrocarbon-rich phase (upper phase), its composition was calculated using literature data of density vs. composition for the binary systems {octane (1) + benzene (2)} and {decane (1) + benzene (2)} at T = 298.15 K [19,21] and experimental data for the systems {octane (1) + benzene (2)} at T = (308.15 and 318.15) K, which are presented in Table S2 (available as Supporting Information). Since the ILs studied are not miscible in the hydrocarbons used in this work, their presence in the aliphatic hydrocarbon-rich phase was neglected; this absence was checked by ^{1}H NMR. Although many authors consider that the determination of the tie–lines ends compositions, especially the composition of the extract phase, by means of the previous determination of the solubility curves by the "cloud point" method using density measurements is not an adequate method over the last years; given that the use of only one physical property leads to the solubility and binodal curves are forced to be coincident. In this work, this method has been used due to the fact that the tie–line ends are close to the binary systems. Additionally, in order to verify the consistency of this method: i) the trend of the iso–lines of densities were determined and they intersect with the binodal curve; in this way, a certain density value corresponds to one tie-line and ii) it was ascertained that the initial mixtures of the LLE (ternary mixtures prepared by weighing within the immiscible region of the ternary system) belong to the corresponding calculated tie–line. Therefore, this method is sound and implies a low level of error in the type of ternary systems presented in this work.

3. Results and discussion

3.1. Experimental LLE data

In order to have an initial glimpse of the solubility of benzene and octane, decane and dodecane in the studied ammonium-based ILs a study of the solubility of the aliphatic hydrocarbons was performed previously to the determination of the LLE of the ternary systems, together with a study of the influence of the temperature in the solubility of benzene and octane in the ionic liquid $[N_{4441}][NTf_2]$. In the case of the study of the solubility of the aliphatic hydrocarbons in $[N_{4111}][NTf_2]$, the obtained experimental solubilities in mole fraction of the corresponding hydrocarbon were: $x_{octane} = 0.039$, $x_{decane} = 0.021$ and $x_{dodecane} = 0.007$. Moreover, the experimental solubilities in mole fraction of the corresponding hydrocarbon were: $x_{octane} = 0.093$ and $x_{dodecane} = 0.019$ when the used ionic liquid was $[N_{4441}][NTf_2]$. These results are in agreement with those previously reported in literature [22,23]. As it can be observed, an increase in the alkyl chain length of the aliphatic hydrocarbon leads to a decrease in their solubility in the ILs. Regarding the solubility of benzene in the studied ILs, the experimental solubilities in mole fraction of benzene were: $x_{benzene} = 0.757$ in $[N_{4111}][NTf_2]$ and $x_{benzene} = 0.858$ in $[N_{4441}][NTf_2]$. The solubility of benzene is much higher than the solubility of aliphatic hydrocarbons in the ILs studied, as it has also been observed by many researchers in other ionic liquid-hydrocarbon binary systems [22,24–26]. Besides, it is possible to observe that all of the studied hydrocarbons are more soluble in $[N_{4441}][NTf_2]$ than in $[N_{4111}][NTf_2]$.

On the other hand, the study of the effect of the temperature in the solubility of benzene and octane in $[N_{4441}][NTf_2]$ was performed. The experimental solubilities in mole fraction of octane were found to be: $x_{octane} = 0.093$, $x_{octane} = 0.119$ and $x_{octane} = 0.131$ at T = (298.15, 308.15 and 318.15) K, respectively. These results show that an increase in the temperature leads to an increase in the solubility of the aliphatic hydrocarbon. On the contrary, the solubility of benzene is practically independent of the increase in the temperature, since the experimental solubilities in mole fraction of benzene were $x_{benzene} = 0.858$, $x_{benzene} = 0.858$ and $x_{benzene} = 0.862$, at T = 298.15, 308.15 and 318.15 K, respectively.

Furthermore, the experimental LLE results for the ternary systems {octane or decane or dodecane (1) + benzene (2) + $[N_{4111}][NTf_2]$ (3)} and {dodecane (1) + benzene (2) + $[N_{4441}][NTf_2]$ (3)} at T = 298.15 K and {octane (1) + benzene (2) + $[N_{4441}][NTf_2]$ (3)} at T = (298.15, 308.15 and 318.15) K are summarized in Table 2 and the corresponding plots are presented in the triangular diagrams shown in Fig. 1, except the ternary diagrams {octane (1) + benzene (2) + $[N_{4441}][NTf_2]$ (3)} at T = (308.15 and 318.15) K which are shown in Fig. S1 (available as Supporting Information).

For characterizing the suitability of the ionic liquids for the extraction of benzene from its mixtures with the studied aliphatic hydrocarbons (octane, decane and dodecane), the solute distribution ratio, β, and the selectivity, S, were calculated according to the following equations:

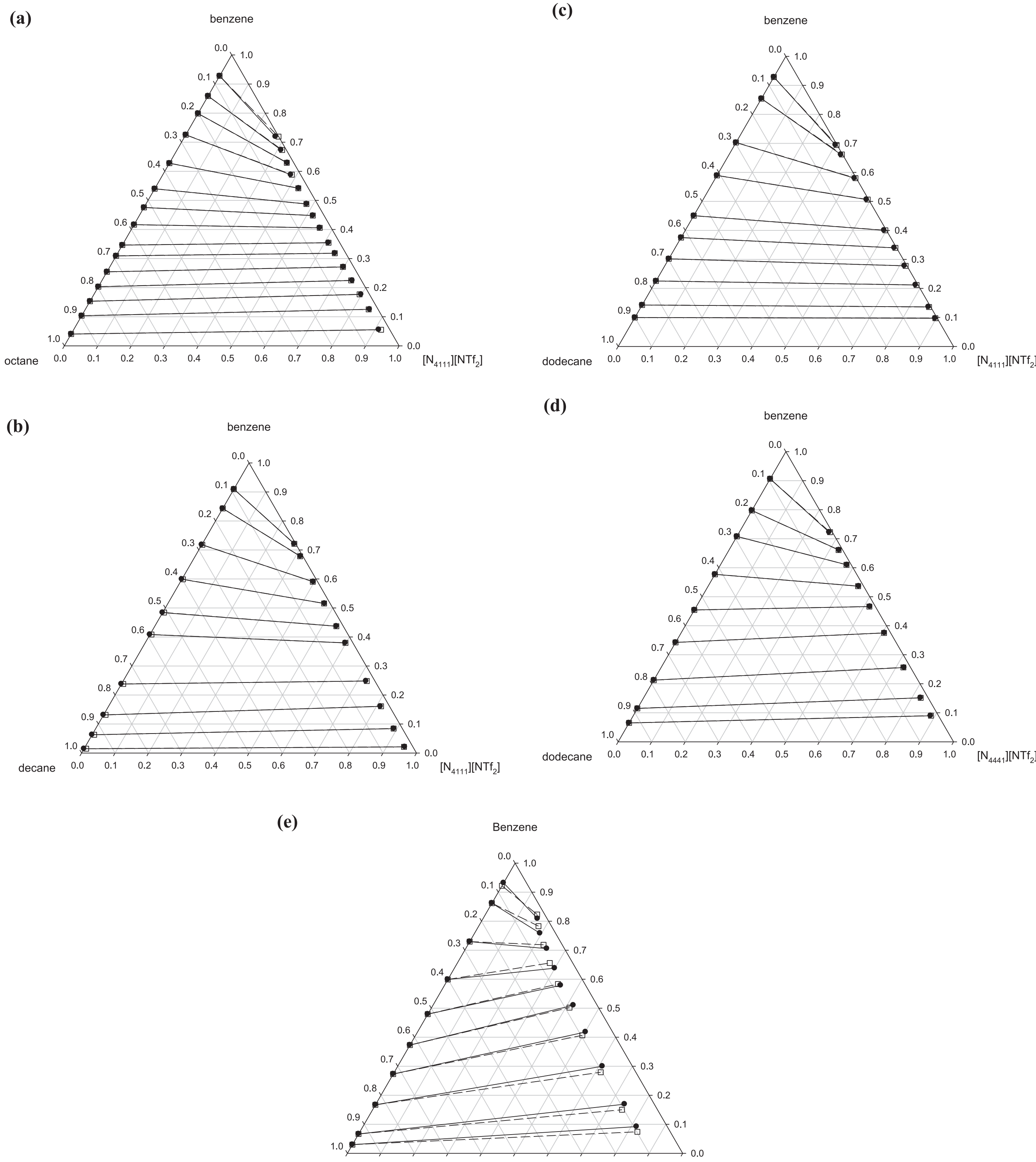

Fig. 1. Experimental and calculated LLE data for the ternary systems: a) {octane (1) + benzene (2) + $[N_{4111}][NTf_2]$ (3)}, b) {decane (1) + benzene (2) + $[N_{4111}][NTf_2]$ (3)}, c) {dodecane (1) + benzene (2) + $[N_{4111}][NTf_2]$ (3)}, d) {dodecane (1) + benzene (2) + $[N_{4441}][NTf_2]$ (3)} and e) {octane (1) + benzene (2) + $[N_{4441}][NTf_2]$ (3)} at $T = 298.15$ K and atmospheric pressure. Solid circles and solid lines represent experimental tie-lines, and empty squares and dashed lines represent calculated data using the NRTL model.

$$\beta = x_2^{\mathrm{II}} / x_2^{\mathrm{I}} \quad \text{or} \quad \beta = w_2^{\mathrm{II}} / w_2^{\mathrm{I}} \tag{1}$$

$$S = x_2^{\mathrm{II}} x_1^{\mathrm{I}} / x_2^{\mathrm{I}} x_1^{\mathrm{II}} \tag{2}$$

where x is the mole fraction and w is the mass fraction, superscripts

I and II indicate the upper (aliphatic hydrocarbon-rich phase) and lower (ionic liquid-rich phase) phase, respectively, and subscripts 1 and 2 refer to the corresponding aliphatic hydrocarbon and benzene, respectively. These parameters are presented in Table 2 and plotted in Fig. 2. The solute distribution ratio is related to the amount of solvent required for the extraction process and to the capacity of the solvent to extract benzene while the selectivity evaluates the efficiency of the solvent used and indicates the ease of extraction of benzene from its mixtures with the studied aliphatic hydrocarbons.

The influence of the temperature in the ternary system {octane (1) + benzene (2) + [N_{4441}][NTf_2] (3)} is shown in Fig. 2. The effect of the temperature on the separation of benzene from octane is rather small, since the values of the solute distribution ratio and the selectivity are quite similar at the three studied temperatures. This result is in concordance with the results obtained for other ternary systems reported in literature [27–29].

Moreover, the influence of the cation of the ionic liquid on the LLE determination can be studied in the ternary systems {octane (1) + benzene (2) + [N_{4111}][NTf_2] or [N_{4441}][NTf_2] (3)} and {dodecane (1) + benzene (2) + [N_{4111}][NTf_2] or [N_{4441}][NTf_2] (3)}. The values of the solute distribution ratio for these systems can be observed in Fig. 2(a). Those ternary systems containing [N_{4111}][NTf_2] as solvent present lower values of β than the ternary systems in which the ionic liquid is [N_{4441}][NTf_2]; this trend is observed for the ternary systems involving octane and dodecane as aliphatic hydrocarbon. In Fig. 2(b), it is observed that the selectivities show the inverse trend to the solute distribution ratios, being the S values of the systems with [N_{4111}][NTf_2] higher than those obtained with the other studied ionic liquid.

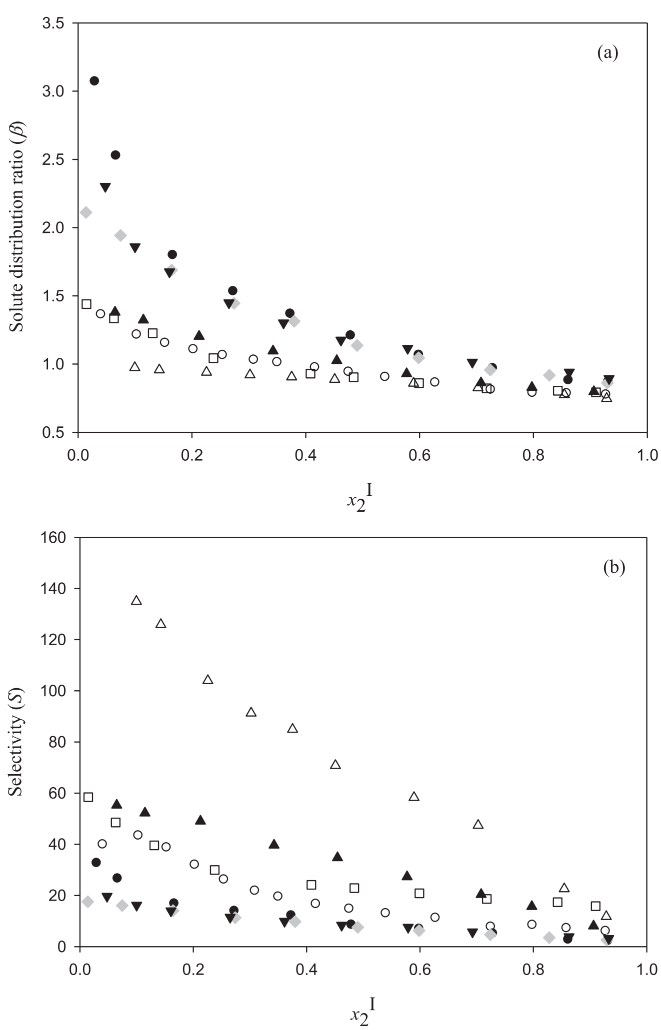

Fig. 2. (a) Solute distribution ratio and (b) selectivity as a function of the mole fraction of benzene in the aliphatic hydrocarbon-rich phase for the ternary systems {octane (1) + benzene (2) + [N_{4111}][NTf_2] (3)} (○), {decane (1) + benzene (2) + [N_{4111}][NTf_2] (3)} (□), {dodecane (1) + benzene (2) + [N_{4111}][NTf_2] (3)} (△), {octane (1) + benzene (2) + [N_{4441}][NTf_2] (3)} (●) and {dodecane (1) + benzene (2) + [N_{4441}][NTf_2] (3)} (▲) at T= 298.15 K and {octane (1) + benzene (2) + [N_{4441}][NTf_2] (3)} (light grey ◆) at T = 308.15 K and {octane (1) + benzene (2) + [N_{4441}][NTf_2] (3)} (▼) at T = 318.15 K and atmospheric pressure.

Regarding the influence of the alkyl chain length of the aliphatic hydrocarbons in the ternary systems {octane or decane or dodecane (1) + benzene (2) + [N_{4111}][NTf_2] (3)} and {octane or dodecane (1) + benzene (2) + [N_{4441}][NTf_2] (3)}, it can be seen that the values of β are higher in the systems containing octane for both studied ILs. Moreover, the selectivities show lower values when the chain length of the aliphatic hydrocarbon decreases.

In the case of the ternary systems with [N_{4111}][NTf_2] as separation agent, the extraction of benzene from decane was also studied. Therefore, the influence of the alkyl chain length of the aliphatic hydrocarbon on the extraction of benzene was analyzed. The values of the solute distribution ratio follow the order: octane ≥ decane > dodecane, whereas the selectivity values decrease when the chain alkyl length of the aliphatic hydrocarbon increases.

Additionally, a comparison between the experimental data at T = 298.15 K and those reported in literature for sulfolane, which is the most industrially used traditional organic solvent, was carried out [30,31]. In order to make a reliable comparison, the solute distribution ratio and the selectivity were calculated in mass fraction instead of mole fraction, since the difference between the molar masses of the studied ILs and sulfolane is relatively high. Besides, it should be mentioned that the selectivities are independent from the mole or mass basis, while the solute distribution ratio calculated using mass fractions give an idea of the quantity of solvent needed for the extraction process. Moreover, although the LLE data of the systems containing decane and dodecane with sulfolane as solvent have been reported in literature at T = 303.15 K [31], the comparison is possible since the effect of the temperature on the LLE is not determining, as it has been previously commented and demonstrated by different authors [27–29].

As it can be observed in Fig. 3(a), the values of β reported for the ternary systems involving sulfolane are higher than those obtained in this work, being the followed order of the β values: sulfolane > [N_{4441}][NTf_2] > [N_{4111}][NTf_2]. Therefore, a lower amount of sulfolane than of the ammonium-based ILs would be required for a separation of benzene from its mixtures with the studied aliphatic hydrocarbons.

By inspection of Fig. 3(b) it can be inferred that the selectivity values calculated for the system {octane (1) + benzene (2) + sulfolane (3)} are higher than those obtained in this work, following the order: sulfolane > [N_{4441}][NTf_2] > [N_{4111}][NTf_2] for $w_2^{\mathrm{I}} \leq 0.50$. When the aliphatic hydrocarbon is changed by decane, the S values obtained for the system involving [N_{4111}][NTf_2] are higher than those for the system with sulfolane for $w_2^{\mathrm{I}} \geq 0.35$. Finally, in the systems containing dodecane, the S values obtained for the systems with the ammonium-based ILs as solvents are higher than the system with sulfolane in the whole range of the concentration of benzene.

3.2. Thermodynamic correlation

All the ternary systems were correlated using the Non-Random Two-liquid (NRTL) model [12]. Although this model was not originally developed for systems involving electrolytes, it is usually

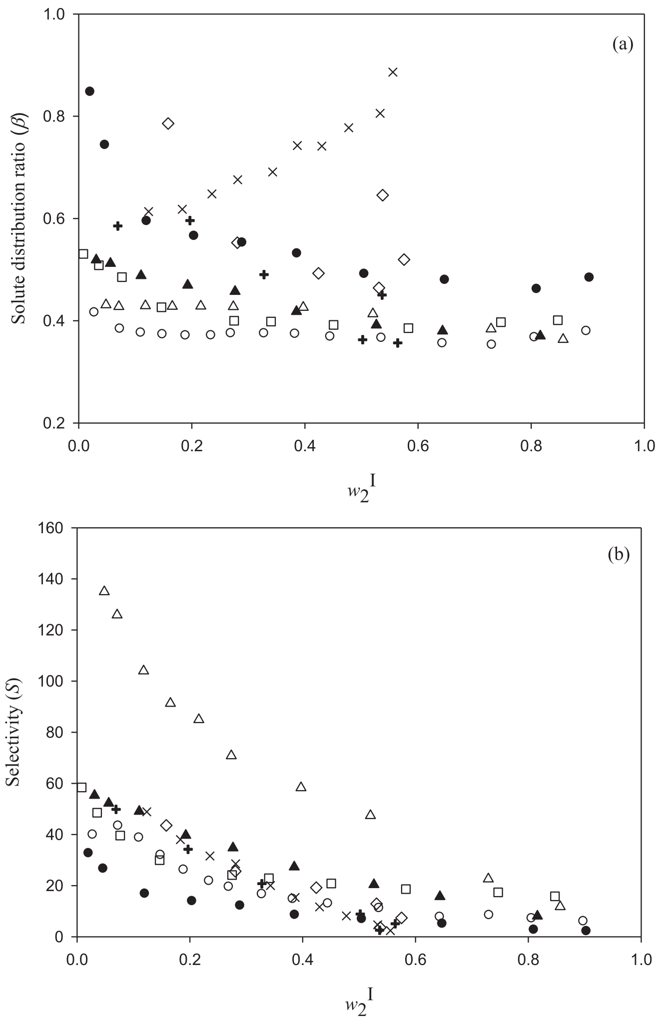

Fig. 3. (a) Solute distribution ratio and (b) selectivity as a function of the mass fraction of benzene in the aliphatic hydrocarbon-rich phase for the ternary systems {octane (1) + benzene (2) + $[N_{4111}][NTf_2]$ (3)} (○), {decane (1) + benzene (2) + $[N_{4111}][NTf_2]$ (3)} (□), {dodecane (1) + benzene (2) + $[N_{4111}][NTf_2]$ (3)} (△), {octane (1) + benzene (2) + $[N_{4441}][NTf_2]$ (3)} (●), {dodecane (1) + benzene (2) + $[N_{4441}][NTf_2]$ (3)} (▲) and {octane (1) + sulfolane (3)} (×) [30] at T = 298.15 K and {decane (1) + benzene (2) + sulfolane (3)} (+) and {dodecane (1) + benzene (2) + sulfolane (3)} (◊) at T = 303.15 K [31].

applied to correlate ternary systems containing ionic liquids with satisfactory results [24,32–35]. In this work the non-randomness parameter, α_{ij}, was fixed to values ranging from 0.05 to 0.4 during the calculations, obtaining the best results when it is set to 0.10.

Table 3 lists the values of the calculated NRTL parameters for all the studied ternary systems: the binary interaction parameters, Δg_{ij}, the non-randomness parameters, α_{ij}, and the root-mean-square deviations of the compositions, σx, which were calculated as follows:

$$\sigma x = \sqrt{100\left(\frac{\sum_{i}^{m}\sum_{j}^{n-1}\left(x_{ij}^{\mathrm{I,exp}} - x_{ij}^{\mathrm{I,cal}}\right)^2 + \left(x_{ij}^{\mathrm{II,exp}} - x_{ij}^{\mathrm{II,cal}}\right)^2}{2mn}\right)} \quad (3)$$

where x is the mole fraction of the component j for the tie-line i, being j = 1 or 2 for the corresponding aliphatic hydrocarbon and benzene, respectively, the superscripts cal and exp denote the calculated and experimental values, respectively, and m and n are the number of the experimental tie-lines and the components in the mixture, respectively.

Table 3
NRTL parameters, Δg_{ij} and α_{ij}, and root-mean-square deviations of the compositions, σx, obtained from ternary LLE data correlation at the studied temperature and atmospheric pressure.

i-j	Δg_{ij}/(kJ mol^{-1})	Δg_{ij}/(kJ mol^{-1})	α_{ij}	σx
{octane (1) + benzene (2) + $[N_{4111}][NTf_2]$ (3)} at T = 298.15 K				
1–2	−9.638	13.35	0.10	0.138
1–3	123.8	8.944		
2–3	110.7	−3.968		
{decane (1) + benzene (2) + $[N_{4111}][NTf_2]$ (3)} at T = 298.15 K				
1–2	−9.342	9.849	0.10	0.057
1–3	10.41	2.501		
2–3	113.7	−4.941		
{dodecane (1) + benzene (2) + $[N_{4111}][NTf_2]$ (3)} at T = 298.15 K				
1–2	−9.523	15.50	0.10	0.117
1–3	21.57	3.294		
2–3	94.33	−9.387		
{dodecane (1) + benzene (2) + $[N_{4441}][NTf_2]$ (3)} at T = 298.15 K				
1–2	−6.348	12.91	0.10	0.069
1–3	61.24	4.482		
2–3	128.2	−0.382		
{octane (1) + benzene (2) + $[N_{4441}][NTf_2]$ (3)} at T = 298.15, 308.15 and 318.15 K				
1–2	−4.315	8.065	0.10	0.799 (298.15 K)
1–3	92.15	3.995		0.391 (308.15 K)
2–3	149.05	−0.710		0.753 (318.15 K)

As it can be inferred from Table 3, the NRTL model describes quite accurately the experimental values of the LLE, since the deviations obtained are quite small. The goodness of the correlation can also be visually confirmed in Fig. 1, where experimental and calculated data are shown.

4. Conclusions

LLE data for the ternary systems {octane (1) + benzene (2) + $[N_{4111}][NTf_2]$ (3)}, {decane (1) + benzene (2) + $[N_{4111}][NTf_2]$ (3)}, {dodecane (1) + benzene (2) + $[N_{4111}][NTf_2]$ (3)} and {dodecane (1) + benzene (2) + $[N_{4441}][NTf_2]$ (3)} at T = 298.15 K and atmospheric pressure were experimentally determined, together with a study of the influence of the temperature on the LLE through the experimental study of the ternary systems {octane (1) + benzene (2) + $[N_{4441}][NTf_2]$ (3)} at T = (298.15, 308.15 and 318.15) K and atmospheric pressure. Solute distribution ratios and selectivities were calculated and compared with those reported in literature for sulfolane. The study of the influence of the temperature on the extraction of benzene from its mixtures with octane shows that the temperature is not determining in the extraction process. All systems showed selectivity values higher than the unity; hence, the studied ILs could be used for extraction processes. The analysis of the influence of the cation on the extraction of benzene shows that the systems containing $[N_{4111}][NTf_2]$ present lower β values than the systems in which the ionic liquid is $[N_{4441}][NTf_2]$, while the selectivity values present the inverse trend. With these results, it can be concluded that a lower amount of the ionic liquid $[N_{4441}][NTf_2]$ than of $[N_{4111}][NTf_2]$ would be necessary in the extraction process. Regarding the influence of the alkyl chain length of the aliphatic hydrocarbons, an increase in the alkyl chain length of the aliphatic hydrocarbon leads to lower β values and higher S values. The highest selectivity values were obtained for the ternary systems containing dodecane. Moreover, considering the comparison of the obtained results and those found in literature for sulfolane as separation agent, the studied ILs present higher selectivities than sulfolane when the aliphatic hydrocarbon is dodecane. Finally, the NRTL model was used to correlate satisfactorily the experimental LLE data for all the studied ternary systems.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at http://dx.doi.org/10.1016/j.fluid.2016.02.006.

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